Memory Notes

Simplified LPDDR3 Simplified edition - JEDEC JESD209-3C

October 3, 2026

Table of Contents

1 Simplified Lpddr3 Index....11

2 Introduction....12

2.1 What LPDDR3 SDRAM is....12

2.2 Family positioning....13

2.3 Key features of LPDDR3....14

2.4 Supplies and voltage ranges....15

2.5 Speed grades....16

2.6 The simplified drill model....17

3 Organization....18

3.1 Banks, rows, columns....18

3.2 Prefetch and IO width....19

3.3 Burst structure....20

3.4 Channels....21

3.5 The simplified drill model....22

4 Acronyms and Conventions....23

4.1 Signal names....24

4.2 Commands....25

4.3 Architecture terms....26

4.4 Power and refresh terms....27

4.5 Conventions....28

5 Power-Up and Initialization....29

5.1 Cold-start sequence....30

5.1.1 Figure 2.1: LPDDR3 Power-Up Initialization Timeline....31
5.2 Reset without power ramp....33
5.3 Power-off sequence....34
5.4 Notes that bite....35
6 Mode Registers....36
6.1 Register map (SDRAM view)....37
6.2 Programming rules....39
6.3 MR0 - Device Information....40
6.4 MR1 - Device Feature 1....41
6.5 MR2 - RL and WL....42
6.6 MR3 - Drive Strength....43
6.7 MR4 - Refresh Rate and Temperature....44
6.8 MR10 - ZQ Calibration Codes....45
6.9 MR11 - ODT Control....46
6.10 MR16 / MR17 - PASR Masks....47
6.11 MR32 / MR40 - DQ Calibration Patterns....48
6.12 MR41 / MR42 / MR48 - CA Training....49
6.13 MR63 - Reset....50
7 Command Truth Tables....51
7.1 Pin conventions....52
7.2 Command truth table....53
7.3 CKE table....55
7.4 State table highlights....57
8 Commands: Rules and Constraints....58

8.1 ACT - Activate..... 59
8.2 RD - Burst read..... 60
8.3 WR - Burst write..... 61
8.4 PRE - Precharge..... 62
8.5 REFab / REFpb..... 63
8.6 Self-refresh entry and exit..... 64
8.7 MRR / MRW..... 65
8.8 Calibration and training..... 66
8.9 ODT..... 67
8.10 Power-down..... 68
8.11 Deep power-down..... 69
8.12 Clock stop and frequency change..... 70
8.13 NOP / Deselect..... 71
9 Walkthrough: AXI Read, Row Already Open..... 72
9.1 Scenario..... 73
9.2 Command trace..... 74
9.3 Why each command..... 75
9.4 AXI view..... 76
9.5 What to notice..... 77
10 Walkthrough: AXI Read, Wrong Row Open..... 78
10.1 Scenario..... 78
10.2 Command trace..... 79
10.3 Why each command..... 80
10.4 AXI view..... 81

10.5 What to notice....82
11 Walkthrough: AXI Write to an Idle Bank....83
11.1 Scenario....83
11.2 Command trace....84
11.3 Why each command....85
11.4 AXI view....86
11.5 What to notice....87
12 Walkthrough: Mixed Reads and Writes....88
12.1 Scenario....88
12.2 Command trace....89
12.3 Why each gap....90
12.4 AXI view....91
12.5 What to notice....92
13 Data Path....93
13.1 Prefetch and burst....93
13.2 Read and write strobes....94
13.3 Read and write latencies....95
13.4 DM - write data mask....97
13.5 What LPDDR3 does not have....98
13.6 Signaling....99
14 On-Die Termination....100
14.1 Enabling and selecting the termination value....101
14.2 Asynchronous ODT timing....102
14.3 ODT during reads....103

14.4 ODT during low-power states and training....104
14.5 ODT states truth table....105
14.6 What LPDDR3 does not have....106
15 Core Timing Parameters....107
15.1 Row-cycle parameters....108
15.2 Activate-rate parameters (cross-bank)....109
15.3 Data-to-precharge parameters....110
15.4 Why these numbers dominate....111
16 Turnaround and Burst Timing....112
16.1 Latencies and CAS-to-CAS....113
16.2 Direction changes....114
16.3 Strobe-level timing (for completeness)....115
16.4 Temperature derating....116
17 Refresh and Self-Refresh Timing....117
17.1 Refresh parameters....117
17.2 Self-refresh....119
17.3 Power-down and deep power-down....120
18 Package and Electrical Notes....121
18.1 Packages....122
18.2 Supplies and signaling....123
18.3 Speed grades....124
18.4 Derating and characterization....125
19 Test Features....126
19.1 MRR as the observation port....127

19.2 DQ calibration (MR32 / MR40)....128
19.3 ZQ calibration (MR10)....129
19.4 CA training (MR41 / MR48 / MR42)....130
19.5 Write leveling (MR2[7])....132
19.6 Temperature sensor (MR4)....133
19.7 What is not here....134

List of Figures

Figure 2.1: LPDDR3 Power-Up Initialization Timeline.....31

List of Tables

No tables in this document.

List of Waveforms

No waveforms in this document.

1 Simplified Lpddr3 Index

Generated: 2026-10-20

2 Introduction

2.1 What LPDDR3 SDRAM is

LPDDR3 SDRAM is the third generation of JEDEC low-power double-data-rate memory, standardized as JESD209-3C. It targets mobile and battery-powered systems, typically in package-on-package (PoP) or discrete FBGA form. The interface runs at 1.2 V and the devices carry a set of power-reduction modes (power-down, self-refresh, deep power-down, per-bank refresh, clock stop) that are not found on desktop DDR3.

Its defining architectural choice is the 10-bit double-data-rate command/address (CA) bus. Instead of a wide SDR address bus with separate RAS_n, CAS_n and WE_n strobes, commands and row/column/bank addresses are packed onto two edges of a single clock cycle on CA0-CA9. CS_n and CKE are the only control inputs sampled on a single clock edge.

2.2 Family positioning

Tech	Prefetch	IO voltage	DLL	Command/ address style
LPDDR2 (JESD209-2)	4n (S4) / 2n (S2)	1.2 V (HSUL_12)	No	10-bit DDR CA bus
LPDDR3 (JESD209-3)	8n	1.2 V (HSUL_12)	No	10-bit DDR CA bus
DDR3 (JESD79-3)	8n	1.5 V (SSTL_15)	Yes	Full parallel SDR CA + RAS/CAS/WE

LPDDR3 is therefore best understood as an evolutionary step from LPDDR2: same CA-bus concept, wider prefetch, faster data rates, and a richer set of training/calibration features.

2.3 Key features of LPDDR3

- 8n prefetch: a column access moves eight words internally per DQ per access; the DDR interface delivers two words per DQ per clock, so the minimum burst occupies four clocks.
- Burst length fixed at BL8. MR1 burst-length encoding is 011B for BL8; all other encodings are reserved. There is no burst chop.
- 8 banks flat, addressed by BA0-BA2. There are no bank groups and no stack IDs.
- x16 and x32 data widths; packages support single- or dual-channel arrangements, but the protocol in this book is per-channel.
- No DLL. Read data timing is given as a broad tDQSCK window relative to the clock; tDQSCK may extend across more than one clock period.
- Read latency (RL) and write latency (WL) are programmed through mode registers (MR0/MR2), not via additive latency.
- No dedicated RESET# pin. Reset is performed with an MRW RESET command.
- MRR and MRW commands replace the single MRS command used by desktop DDR3.
- Training and calibration: CA training, write leveling, DQ calibration, and ZQ calibration are all supported.
- Asynchronous ODT: DQ-bus termination follows the ODT pin and the MR11 setting, and is turned off during reads, power-down, self-refresh, deep power-down and CA training.
- Deep power-down (DPD): lowest-power state; array contents are lost and exit requires full re-initialization.
- Per-bank refresh (REFpb) and all-bank refresh (REFab) both exist.
- Temperature sensor readable via MR4, used for refresh-rate scaling and AC timing de-rating decisions.
- Clock stop supported during CKE LOW and CKE HIGH under defined conditions.

2.4 Supplies and voltage ranges

Supply	Nominal	Function
VDD1	1.80 V	Core power
VDD2	1.20 V	Core power
VDDCA	1.20 V	CA/CKE/CS_n/CK input-buffer power
VDDQ	1.20 V	DQ/DQS/DM I/O power
VREF(CA)	VDDCA/2	Reference for CA, CKE, CS_n, CK
VREF(DQ)	VDDQ/2	Reference for DQ inputs

2.5 Speed grades

Data-rate grades reach 1333, 1600, 1866 and 2133 MT/s at the high end (Table 64); lower data rates are also supported. Core timings (tRCD, tRP, tRAS) are specified in nanoseconds, while RL and WL are given in clock counts that depend on data rate (Table 63).

2.6 The simplified drill model

Real parts have thousands of rows and hundreds of columns. All examples in this book use a deliberately tiny geometry so bank state is obvious at a glance:

- Banks: B0-B7 (eight, as on 1 Gb and larger parts)
- Rows: R0-R7 per bank
- Columns: C0-C7 per row

This model is used by the ddr_drills training app. Every command trace, walkthrough and timing example in later chapters uses it. Real address maps are far larger; the scheduling rules are identical.

Source: JESD209-3C sections 1, 2.4, 3, Table 64

3 Organization

3.1 Banks, rows, columns

An LPDDR3 device is organized as eight independent banks of DRAM arrays. Each bank has its own sense-amplifier row buffer, so up to eight rows can be open at the same time (one per bank). The bank count is always eight; the spec does not define 4-bank LPDDR3 SDRAM devices.

Address geometry per density (from the spec addressing table):

Density	Banks	Rows (x16)	Columns (x16)	tREFI (us, <=85 C)
1 Gb	8	R0-R12	C0-C9	7.8
2 Gb	8	R0-R13	C0-C9	3.9
4 Gb	8	R0-R13	C0-C10	3.9
6 Gb	8	R0-R14*	C0-C10	3.9
8 Gb	8	R0-R14*	C0-C10	3.9
12 Gb	8	R0-R14*	C0-C11	3.9
16 Gb	8	R0-R14*	C0-C11	3.9
32 Gb	8	TBD	TBD	TBD

*Addresses with R13 and R14 both high have no memory; an ACT to that space is ignored.

x32 parts lose one column address bit relative to x16; x16 parts expose one more column bit than x32 at the same density.

Addressing rules:

- The least-significant column bit C0 is implied zero and does not appear on the CA bus. Column addresses on the bus start at C1.
- REFpb can only target a bank that is currently idle and precharged.

3.2 Prefetch and IO width

- 8n prefetch: a column access fetches eight words per DQ internally. The DDR interface delivers two words per DQ per clock, so a minimum burst occupies four clocks. This is why tCCD = 4 clocks on LPDDR3.
- Data bus widths are x16 and x32. Each byte lane has its own differential strobe (DQS_t/DQS_c) and its own DM mask pin: one DQS pair and one DM per 8 DQs.

3.3 Burst structure

Accesses are burst-oriented: each Read or Write command transfers BL words per DQ, starting at the issued column and following the programmed burst order. BL is fixed at 8. Burst addressing follows the standard sequential pattern driven by the low-order column bits.

3.4 Channels

Multi-channel parts exist at the package level: a PoP or discrete package can carry two fully independent LPDDR3 channels, each with its own CA bus, clock, CKE and CS_n. Channels share nothing but the package; the protocol in this book is per-channel.

3.5 The simplified drill model

Per the Introduction, all examples use eight banks (B0-B7), eight rows (R0-R7) and eight columns (C0-C7). Real parts are far larger; the small model exists to make bank/row state obvious in command traces.

Source: JESD209-3C sections 2.1-2.3, 3.1

4 Acronyms and Conventions

Terms used precisely in this book.

4.1 Signal names

Term	Meaning
CK_t / CK_c	Differential clock. Positive edge = CK_t rising through CK_c falling
CKE	Clock enable. SDR, sampled on the positive clock edge; gates all power states
CS_n	Chip select. SDR, sampled on the positive clock edge; contributes to the command code
CA0-CA9	10-bit DDR command/address bus; sampled on both clock edges
CA / CATR	Command/address bus / CA training mode entered via MR41/MR42/MR48
CAxr / CAxf	CA bit x on the rising (r) / falling (f) edge of the clock
DQ	Bidirectional data
DQS_t / DQS_c	Differential data strobe, one pair per byte lane; edge-aligned on reads, center-aligned on writes
DM	Data mask, one per byte lane; masks write data when sampled high
ODT	Asynchronous on-die termination control pin for the DQ bus
ZQ	Pin used as reference for output drive-strength calibration

4.2 Commands

Term	Meaning
ACT	Activate (open a row in a bank)
RD / WR	Burst read / burst write
PRE	Precharge (close a row); per-bank or all-bank via the AB flag
REFab / REFpb	Refresh, all banks / one per-bank-refresh bank
MRR / MRW	Mode register read / mode register write
NOP	No operation
SREF / PD / DPD	Self-refresh, power-down, deep power-down

4.3 Architecture terms

Term	Meaning
Prefetch (8n)	Words moved internally per column access; LPDDR3 = 8n
BL	Burst length; fixed at 8 for LPDDR3
RL / WL	Read latency / write latency, in clocks, programmed in MR0/MR2
nWR	Write recovery in clocks for auto-precharge, programmed in MR1
AP	Auto-precharge flag (CA0f of a RD/WR command)
AB	All-bank flag (CA4r of a PRE command)
RU{}	Round-up function; RU{x} = smallest integer >= x
CA training	Training mode to center CA inputs; entered/exited via MR41/MR42
Write leveling	Training mode to align DQS with the clock; entered/exited via MRW
DQ calibration	MRR reads from MR32/MR40 return predefined calibration patterns

4.4 Power and refresh terms

Term	Meaning
PASR	Partial array self-refresh: refresh only masked banks/segments in self-refresh
RM	Refresh multiplier from MR4; tREFIM = RM x tREFI
DAI	Device auto-initialization status bit (MR0 OP0)
TUF	Temperature update flag (MR4 OP7)
DPD	Deep power-down; array contents lost, full re-init required on exit
HSUL_12	High-speed unterminated logic, 1.2 V: the LPDDR3 IO standard
tREFW / tREFI / tREFIM	Refresh window / average refresh interval / multiplied refresh interval
tRFCab / tRFCpb	Refresh cycle time, all-bank / per-bank
RZQ	240 ohm reference resistor for output driver and ODT calibration

4.5 Conventions

- H/L in truth tables are logic high/low at the sampled edge; X is “don’t care” but must still be a valid logic level.
- Timing symbols are written as plain text: tRCD, tWTR, tREFIpb.
- Anything the spec leaves unspecified is illegal; after an illegal event the part has to be reset through a full power-down/re-init cycle.

Source: JESD209-3C sections 2.4 (Table 2), 3.4.1, 4.8, 4.9.1, 4.12

5 Power-Up and Initialization

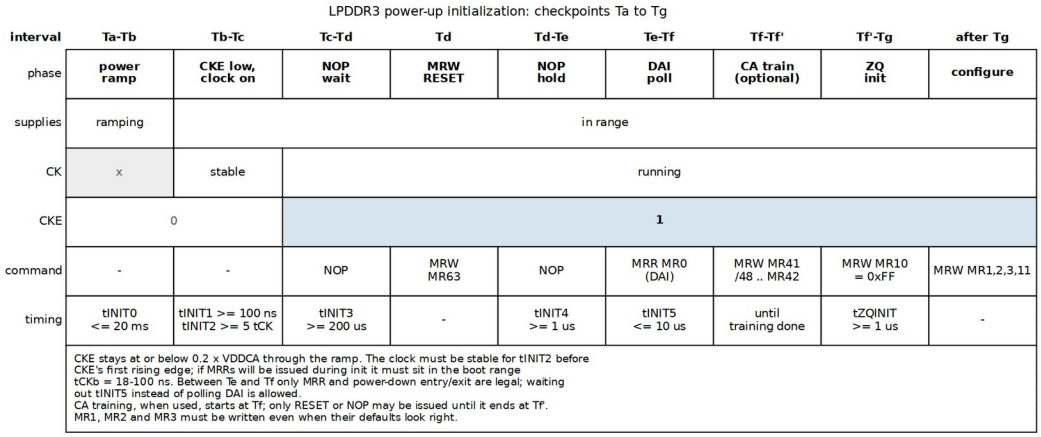
LPDDR3 must be brought up with a fixed sequence; any other path is undefined. Reset is command-based only - the spec has no dedicated SDRAM RESET# pin (the RST_n ball in PoP ballouts is an eMMC signal), so the reset event is an MRW to MR63.

5.1 Cold-start sequence

1. **Power ramp (Ta -> Tb).** Apply supplies while CKE is held LOW (<= 0.2 x VDDCA). Ta is when any supply first reaches 300 mV; Tb is when every supply and VREF is inside its operating range. The ramp (tINIT0) must finish within 20 ms. During the ramp: VDD1 must stay within 200 mV above VDD2; VDD1 and VDD2 must stay within 200 mV above VDDCA and VDDQ; VREF must be below every supply; and VSS/VSSQ/VSSCA must be within 100 mV of each other. DQ, DM, DQS_t and DQS_c must stay between VSSQ and VDDQ; CK_t, CK_c, CS_n and CA must stay between VSSCA and VDDCA, to avoid latch-up. Device outputs are High-Z while CKE is LOW.
2. **Clock and CKE (Tb -> Tc).** Keep CKE LOW for at least tINIT1 = 100 ns after Tb. The clock must be stable for at least tINIT2 = 5 tCK before CKE's first LOW-to-HIGH transition. If any MRR will be issued during init, the clock period must be inside the boot window tCKb = 18-100 ns. CKE, CS_n and CA must meet setup/hold from the first rising edge onward. The ODT input may be undefined until tIS before CKE is registered HIGH, then must be held static LOW or HIGH until initialization is complete (including tZQINIT).
3. **NOP wait (Tc -> Td).** With CKE HIGH, issue NOP for at least tINIT3 = 200 us.
4. **MRW RESET (Td).** Issue an MRW RESET (MR63). An optional PRECHARGE ALL may precede it. Keep CKE HIGH with NOPs for at least tINIT4 = 1 us; only NOPs are allowed during this time.
5. **DAI polling (Te -> Tf).** Only MRR and power-down entry/exit are legal now. Poll MR0's DAI bit, or simply wait tINIT5(max) = 10 us. The device is idle once DAI indicates completion or when tINIT5(max) expires. This is also the normal point to read MR8 (and MR5-MR7) to discover I/O width, density, manufacturer and revision.
6. **ZQ calibration / CA training (Tf -> Tg).** If CA training is required, start it at Tf using MR41 (or MR48 for the second mapping) and exit with MR42; no CA commands other than RESET or NOP may be issued until training finishes (Tf'). Then issue the MRW ZQ initialization calibration command (MR10, code 0xFF). If CA training is not needed, ZQ initialization may start at Tf directly. On a shared ZQ bus, do not overlap ZQ calibrations across devices. The device is ready for normal operation after tZQINIT = 1 us.
7. **Configure (after Tg).** MRW the operating registers: MR1 (burst length, nWR), MR2 (RL/WL, write-leveling enable, WL set), MR3 (drive strength)

and MR11 (DQ ODT). The device is now idle and accepts any valid command.

5.1.1 Figure 2.1: LPDDR3 Power-Up Initialization Timeline



LPDDR3 Power-Up Initialization Timeline

Parameter	Min	Max	Unit	Meaning
tINIT0	-	20	ms	longest allowed supply ramp
tINIT1	100	-	ns	CKE must stay LOW after supplies are valid
tINIT2	5	-	tCK	stable clock before CKE rises
tINIT3	200	-	us	NOP wait after CKE goes HIGH
tINIT4	1	-	us	NOP wait after MRW RESET
tINIT5	-	10	us	maximum

Parameter	Min	Max	Unit	Meaning
				device auto-initialization time
tZQINIT	1	-	us	ZQ initialization calibration
tCKb	18	100	ns	boot clock period for MRR during init

5.2 Reset without power ramp

An MRW RESET issued later (with all banks idle) restarts initialization at Td: mode registers return to defaults, wait tINIT4, poll DAI or wait tINIT5, redo ZQ initialization, then reconfigure MR1/MR2/MR3/MR11. SDRAM contents are undefined after MRW RESET.

5.3 Power-off sequence

Controlled power-off is the ramp rules in reverse. Hold CKE LOW (<= 0.2 x VDDCA); keep all other inputs between VILmin and VIHmax; outputs remain High-Z. DQ, DM, DQS_t and DQS_c must remain between VSSQ and VDDQ; CK_t, CK_c, CS_n and CA must remain between VSSCA and VDDCA. Tx is when any supply first drops below its minimum; Tz is when all supplies are below 300 mV. Between Tx and Tz the same supply-order inequalities apply as during ramp, and VSS pins must stay within 100 mV. The fall from Tx to Tz must not exceed tPOFF = 2 s.

Uncontrolled power-off is tolerated only with tighter limits: at Tx all supplies must turn off and supply current capacity must reach zero (except residual charge); VDD1 and VDD2 must fall with slope <= 0.5 V/us between Tx and Tz; and this may occur at most 400 times over the device's life.

5.4 Notes that bite

- Mode-register defaults are valid only after MRW RESET and DAI completion; before that they are undefined.
- MR1, MR2 and MR3 must be written even if the default values look acceptable.
- Boot-frequency AC relaxations (tDQSCKb, etc.) apply until the device is fully configured.
- Deep power-down exit requires the full power-up initialization sequence from step 3.

Source: JESD209-3C sections 3.3.1, 3.3.2, 4.11, 4.11.1.1

6 Mode Registers

LPDDR3 replaces the MRS/EMRS encoding of desktop DDR with two dedicated commands: MRW (write) and MRR (read). Both carry an 8-bit register address (MA0-MA7) on the CA bus, and each register is 8 bits wide (OP0-OP7). MRW is allowed only when all banks are idle; MRR may be issued from the all-banks-idle or banks-active state.

6.1 Register map (SDRAM view)

MR	Name	Access	Contents
MR0	Device Info	R	DAI, RZQI, WL Set B support, RL3 option support
MR1	Device Feature 1	W	BL, nWR
MR2	Device Feature 2	W	RL/WL, nWRE, WL set select, WR leveling enable
MR3	I/O Config 1	W	DS (drive strength)
MR4	Refresh Rate	R	Temperature-compensated refresh multiplier, TUF
MR5	Basic Config 1	R	Manufacturer ID
MR6	Basic Config 2	R	Revision ID1
MR7	Basic Config 3	R	Revision ID2
MR8	Basic Config 4	R	I/O width, density, type
MR9	Test Mode	W	Vendor-specific test mode
MR10	IO Calibration	W	ZQ calibration codes
MR11	ODT Control	W	DQ ODT value, power-down ODT behavior
MR16	PASR Bank Mask	W	Per-bank self-refresh mask
MR17	PASR Segment Mask	W	Per-segment self-refresh mask
MR32	DQ Calibration Pattern A	R	Read-only training pattern
MR40	DQ Calibration	R	Read-only

MR	Name	Access	Contents
	Pattern B		training pattern
MR41	CA Training 1	W	Enter CA training
MR42	CA Training 2	W	Exit CA training
MR48	CA Training 3	W	Switch CA-to-DQ mapping during CA training
MR63	Reset	W	Any write triggers MRW RESET

Rules: RFU bits are written 0 and read as 0; reserved registers must not be written; writes to read-only registers have no effect.

6.2 Programming rules

- MRW can only be issued when all banks are idle.
- The MRW command period is tMRW = 10 tCK; only NOP is allowed during tMRW.
- The delay from MRW registration to the next valid non-MRW command is tMRD = max(14 ns, 10 nCK).
- The MRR command period is tMRR = 4 tCK; only NOP is allowed during tMRR.
- MRR data is valid only on DQ[7:0] on the first beat; the remaining beats are undefined.
- Minimum MRR-to-MRW latency = RL + RU(tDQSCKmax/tCK) + BL/2 + 1 clock cycles.
- Minimum MRR-to-write latency = RL + RU(tDQSCKmax/tCK) + BL/2 + 1 - WL clock cycles.

6.3 MR0 - Device Information

Field	Bits	Meaning
DAI	OP0	0 = auto-initialization complete; 1 = still in progress
RZQI	OP4:3	ZQ-pin self-test result (optional)
WL Set B support	OP6	1 = device supports WL Set B
RL3 option support	OP7	1 = device supports RL=3/nWR=3/WL=1 at <= 166 MHz

6.4 MR1 - Device Feature 1

Field	Bits	Encodings
BL	OP2:0	011 = BL8 (default); all others reserved
nWR (nWRE=0)	OP7:5	001=3 (optional), 100=6, 110=8, 111=9
nWR (nWRE=1)	OP7:5	000=10 (default), 001=11, 010=12, 100=14, 110=16

Program nWR to RU(tWR/tCK); it sets when the internal precharge starts for a write with auto-precharge.

6.5 MR2 - RL and WL

OP3:0	Set A (OP6=0)	Set B (OP6=1)
0001	RL=3 / WL=1	RL=3 / WL=1
0100	RL=6 / WL=3	RL=6 / WL=3
0110	RL=8 / WL=4	RL=8 / WL=4
0111	RL=9 / WL=5	RL=9 / WL=5
1000	RL=10 / WL=6 (default)	RL=10 / WL=8
1001	RL=11 / WL=6	RL=11 / WL=9
1010	RL=12 / WL=6	RL=12 / WL=9
1100	RL=14 / WL=8	RL=14 / WL=11
1110	RL=16 / WL=8	RL=16 / WL=13

OP4 (nWRE) selects the nWR range; OP6 selects WL Set A or B; OP7 enables write leveling.

6.6 MR3 - Drive Strength

OP3:0	Nominal impedance
0001	34.3 ohm pull-down / pull-up
0010	40 ohm pull-down / pull-up (default)
0011	48 ohm pull-down / pull-up
0100	reserved for 60 ohm
0110	reserved for 80 ohm
1001	34.3 ohm pull-down / 40 ohm pull-up
1010	40 ohm pull-down / 48 ohm pull-up
1011	34.3 ohm pull-down / 48 ohm pull-up

6.7 MR4 - Refresh Rate and Temperature

- OP2:0 reports the refresh multiplier (RM) recommended for the current temperature: 4x, 2x, 1x (<= 85 C), 0.5x, 0.25x without AC de-rating, or 0.25x with AC de-rating. Codes 000 and 111 mean the temperature operating limit is exceeded.
- OP7 (TUF) is set when OP2:0 has changed since the last MR4 read; reading MR4 clears TUF.
- When the de-rating code is selected, add 1.875 ns to tRCD, tRC, tRAS, tRP and tRRD, and apply the tDQSCK de-rating from the AC table.

6.8 MR10 - ZQ Calibration Codes

Code	Command	Latency
0xFF	Initialization calibration	tZQINIT = 1 us
0xAB	Long calibration	tZQCL = 360 ns
0x56	Short calibration	tZQCS = 90 ns
0xC3	ZQ reset (default calibration)	tZQRESET = 50 ns

If ZQ is tied to VDDCA, the device uses default calibration and ignores ZQ commands.

6.9 MR11 - ODT Control

Field	Bits	Encodings
DQ ODT	OP1:0	00 = disable (default), 01 = RZQ/4, 10 = RZQ/2, 11 = RZQ/1
PD Control	OP2	0 = ODT disabled during power-down (default), 1 = ODT enabled during power-down

RZQ/4 is required for LPDDR3-1866 and -2133 and optional for lower speed grades.

6.10 MR16 / MR17 - PASR Masks

- MR16 OP7:0 is a per-bank mask for 8-bank devices; a 1 blocks refresh to that bank.
- MR17 OP7:0 is a per-segment mask; a 1 blocks refresh to that segment of row space in every bank.

Masked regions lose data in self-refresh.

6.11 MR32 / MR40 - DQ Calibration Patterns

MRR to MR32 returns Pattern A; MRR to MR40 returns Pattern B. These read-only patterns are used for DQ calibration.

6.12 MR41 / MR42 / MR48 - CA Training

- MR41 enters CA training.
- MR48 changes the CA-to-DQ mapping so the remaining CA pins (CA4, CA9) can be calibrated.
- MR42 exits CA training.

MR41 and MR42 use special encodings whose rising-edge and falling-edge values are identical, so the device recognizes them even before CA timing has been adjusted.

6.13 MR63 - Reset

Any MRW to MR63 performs a device reset. The operand 0xFC is recommended before CA training because it keeps the CA bus stable across the rising and falling clock edges.

Source: JESD209-3C sections 3.4.1, 4.10, 4.11, 4.11.2, 4.11.3

7 Command Truth Tables

LPDDR3 uses a flat, 8-bank topology with no bank groups and no stack IDs. Commands and addresses travel multiplexed over the 10-bit CA bus on both clock edges; a single command spans one full clock cycle. A hardware RESET# pin is also provided for full device reset.

7.1 Pin conventions

- CKE(n-1) and CKE(n) are sampled at successive rising edges; CS_n is sampled at the rising edge.
- CAxr is CA bit x on the rising edge; CAxf is the same bit on the falling edge.
- H/L are valid logic levels; X means the pin is driven but its value does not matter.

7.2 Command truth table

Command	CKE(n-1)	CKE(n)	CS_n	CA0r	CA1r	CA2r	CA3r	Falling edge payload
MRW	H	H	L	L	L	L	L	MA6 MA7, OP0-OP7
MRR	H	H	L	L	L	L	H	MA6 MA7
REFpb	H	H	L	L	L	H	L	X
REFab	H	H	L	L	L	H	H	X
Enter self-refresh	H	L	L	L	L	H	X	X
ACT	H	H	L	L	H	R8	R9	R0-R7, R13, R14
WR	H	H	L	H	L	L	X	AP, C3-C11
RD	H	H	L	H	L	H	X	AP, C3-C11
PRE (per-bank/all)	H	H	L	H	H	L	H	X
Enter DPD	H	L	L	H	H	L	H	X
NOP	H	H	L	H	H	H	X	X
Deselect (NOP)	H	H	H	X	X	X	X	X
Enter power-	H	L	H	X	X	X	X	X

Command	CKE(n-1)	CKE(n)	CS_n	CA0r	CA1r	CA2r	CA3r	Falling edge payload
down								
Exit PD/SREF/DPD	L	H	H	X	X	X	X	X
Maintain PD/SREF/DPD	L	L	X	X	X	X	X	X

Notes that load-bear:

- ACT: CA4r-CA6r carry R10-R12; CA7r-CA9r select BA0-BA2; the falling edge carries R0-R7, then R13-R14.
- RD/WR: CA5r-CA6r carry C1-C2, CA7r-CA9r select the bank; CA0f is AP and CA1f-CA9f carry C3-C11. C0 is implied zero and never transmitted.
- PRE: CA4r = AB (1 = precharge all banks); when AB = 0, CA7r-CA9r select the bank.
- MRW/MRR: CA4r-CA9r carry MA0-MA5; CA0f = MA6 and CA1f = MA7.
- Self-refresh and DPD exits use an asynchronous CKE rise; power-down exit is synchronous.
- NOP can be issued either by driving CS_n high or by driving CS_n low with CA0r-CA2r high.

7.3 CKE table

Current state	CKE(n-1)	CKE(n)	CS_n	Command	Next state
Active power-down	L	L	X	X	Active power-down
Active power-down	L	H	H	NOP	Active (wait tXP)
Idle power-down	L	L	X	X	Idle power-down
Idle power-down	L	H	H	NOP	Idle (wait tXP)
Resetting power-down	L	L	X	X	Resetting power-down
Resetting power-down	L	H	H	NOP	Idle if tINIT5 elapsed, else Resetting
Self refresh	L	L	X	X	Self refresh
Self refresh	L	H	H	NOP	Idle (wait tXSR)
Deep power-down	L	L	X	X	Deep power-down
Deep power-down	L	H	H	NOP	Power-on (re-init)
Banks active	H	L	H	NOP	Active power-down
All banks idle	H	L	H	NOP	Idle power-down

Current state	CKE(n-1)	CKE(n)	CS_n	Command	Next state
All banks idle	H	L	L	REF code	Self refresh
All banks idle	H	L	L	PRE code	Deep power-down
Any (CKE high)	H	H	-	per command table	-

At least two clock transitions must occur while tXP or tXSR elapses. Sequences not shown are illegal.

7.4 State table highlights

Current state of bank n	Command to bank n	Next state
Idle	ACT	Row active (after tRCD)
Idle	REFpb/REFab	Refreshing
Idle	MRW/MRR/Reset/ PRE	As defined
Row active	RD/WR	Reading/Writing
Row active	PRE	Precharging -> Idle
Row active	MRR	Active MR reading
Reading	RD/WR (after current burst)	Reading/Writing
Writing	WR/RD (after current burst)	Writing/Reading
Refreshing	NOP only	Idle after tRFC
MR writing	NOP only	Idle after tMRW

A command other than NOP must not interrupt an in-flight burst, refresh interval, or MRW window; NOPs are required on every rising edge during those intervals.

Source: JESD209-3C sections 4.1, 4.7, 4.8, 4.9, 4.10-4.11, 4.13-4.17

8 Commands: Rules and Constraints

LPDDR3 has no Mode Register Set command; mode-register updates use MRW and reads use MRR. There is no DLL, no additive latency, no burst terminate, and no burst chop. BL8 is the only burst length.

8.1 ACT - Activate

Opens row R in bank B so column accesses can follow.

- Encoding: CS_n low, CA0r low, CA1r high; CA2r-CA3r carry R8-R9, CA4r-CA6r carry R10-R12, and CA7r-CA9r select the bank; the falling edge carries R0-R7 plus R13-R14.
- RD or WR to the same bank may follow after tRCD.
- ACT to a different bank needs tRRD; ACT to the same bank again needs a PRE first and must satisfy tRAS (ACT-to-PRE) and tRC (ACT-to-ACT).
- On 8-bank devices, no more than four ACT or REFpb operations may fall inside a rolling tFAW window.

Key timing:

Symbol	Rule
tRRD	max(10 ns, 2 nCK) between ACTs to different banks
tFAW	max(50 ns, 8 nCK) rolling window for four bank activations
tRAS	min = max(42 ns, 3 nCK); max = min(70.2 us, 9 x RM x tREFI)
tRC	tRAS + tRPab (with PREab) or tRAS + tRPpb (with PREpb)

8.2 RD - Burst read

Starts a fixed BL8 read burst from the open row.

- Encoding: CS_n low, CA0r high, CA1r low, CA2r high; CA5r-CA6r = C1-C2, CA7r-CA9r = bank, and the falling edge carries AP plus C3-C11.
- RL is programmed in the mode registers. Data is valid after RL x tCK + tDQSCK + tDQSQ from the command edge. Because there is no DLL, tDQSCK spans a wide min/max range.
- Back-to-back reads to any bank are spaced by tCCD = 4 clocks.
- A PRE to the same bank may follow after BL/2 + max(4, RU(tRTP/tCK)) - 4 clocks, and tRAS must also be satisfied.
- **tRTW (book symbol):** the earliest WR after a RD is RL + RU(tDQSCKmax/tCK) + BL/2 + 1 - WL clocks after the RD command.

Key timing:

Symbol	Rule
tRTP	max(7.5 ns, 4 nCK) analog delay from read to PRE

8.3 WR - Burst write

Starts a fixed BL8 write burst.

- Encoding: CS_n low, CA0r high, CA1r low, CA2r low; column, bank, and AP mapping matches RD.
- WL is programmed in the mode registers. Write data is expected WL x tCK + tDQSS after the command edge. The DM pin masks its byte on any beat of the burst.
- Back-to-back writes are spaced by tCCD = 4 clocks.
- A PRE to the same bank may follow after WL + BL/2 + RU(tWR/tCK) + 1 clocks.
- The earliest RD after a WR is WL + 1 + BL/2 + RU(tWTR/tCK) clocks after the WR command.

Key timing:

Symbol	Rule
tWR	max(15 ns, 4 nCK) from last write data to PRE
tWTR	max(7.5 ns, 4 nCK) from last write data to RD

8.4 PRE - Precharge

Closes the open row of one bank or all banks.

- Encoding: CS_n low, CA0r high, CA1r high, CA2r low, CA3r high; CA4r = AB (1 = all banks); when AB = 0, CA7r-CA9r select the bank.
- After a single-bank PRE wait tRPpb; after PREab wait tRPab. tRPab is longer than tRPpb on 8-bank devices.
- PRE must not violate tRAS after ACT, tRTP after RD, or tWR after WR.

8.5 REFab / REFpb

Refreshes rows in the storage array.

- Encoding: CS_n low, CA0r-CA2r = L L H; CA3r = H for all banks, L for per bank.
- REFab refreshes every bank and requires all banks idle; after it wait tRFCab before any ACT or further refresh.
- REFpb refreshes one bank chosen by the device's internal 0-1-…-7 counter; the target bank must be idle. After REFpb wait tRFCpb before an ACT or refresh to that bank; only tRRD is needed before an ACT to a different bank.
- The controller must track the REFpb counter; it resets after device reset, self-refresh exit, and every REFab.
- All-bank refreshes are normally issued every tREFIM = RM x tREFI, where RM comes from MR4. Up to eight refreshes may be postponed, bounding the maximum gap to 9 x tREFIM.

8.6 Self-refresh entry and exit

- Entry: CKE falls while the REF code is on the bus and all banks are idle. Two NOPs are required after CKE goes low (tCPDED). The clock may be stopped or changed tCPDED after entry, but must be stable for at least two cycles before exit.
- Minimum stay is tCKESR.
- Exit: CKE rises asynchronously; issue NOPs until tXSR = max(tRFCab + 10 ns, 2 nCK) expires. At least one refresh (one REFab or eight REFpb) must be issued before re-entering self refresh.
- **PASR:** MR16 masks whole banks and MR17 masks eight segments per bank during self refresh. A region is retained only when both its bank mask bit and its segment mask bit are unmasked.

Region	Mask control	Retained in self refresh when
Whole bank n	MR16 bit n	bit n = 0
Segment s of every bank	MR17 bit s	bit s = 0
Bank n, segment s	MR16 bit n and MR17 bit s	both bits = 0

8.7 MRR / MRW

- MRR encoding: CS_n low, CA0r-CA3r = L L L H; register address on CA4r-CA9r and CA0f-CA1f. Burst length is fixed at 8; valid mode-register data appears on DQ0-DQ7 on the first beat, while later beats are undefined (DQ calibration patterns MR32/MR40 are the exception). DQS toggles for the full burst. tMRR = 4 clocks between MRR commands; only NOPs are allowed during tMRR.
- After idle power-down exit, wait an additional tMRRI = tRCD before MRR.
- MRW encoding: CS_n low, CA0r-CA3r = L L L L; register address on CA4r-CA9r and CA0f-CA1f, data on CA2f-CA9f. MRW is allowed only when all banks are idle; tMRW = 10 clocks, only NOPs allowed. MRW to MR63 triggers device reset and leaves array data undefined.

8.8 Calibration and training

- **ZQ calibration:** initiated by MRW. ZQINIT runs at init; ZQCL is a long recal; ZQCS is a short periodic cal; ZQRESET returns to default. All banks must be idle, ODT disabled, and the DQ bus quiet during ZQINIT/ZQCL/ZQCS.
- **CA training:** enter via MRW to MR41, calibrate CA0-CA3/CA5-CA8, switch CA-to-DQ mapping via MR48, calibrate CA4/CA9, then exit via MR42.
- **Write leveling:** enter by setting MR2[7] high. The DRAM asynchronously feeds back the CK sample on DQ for each DQS group; the controller delays DQS until the 0-to-1 transition is found. Only NOPs or the exit MRW are allowed.
- **DQ calibration:** MRR to MR32 returns Pattern A; MRR to MR40 returns Pattern B on the designated DQ pins.

8.9 ODT

LPDDR3 termination is asynchronous, controlled by the ODT pin and enabled by MR11[1:0]. ODT is automatically disabled during reads, self refresh, deep power-down, and CA training; it may optionally stay on during power-down via MR11[2].

8.10 Power-down

- Entry: CKE falls with CS_n high (NOP/Deselect). Active power-down has at least one row open; idle power-down has all banks idle. No refreshes occur, so dwell time is bounded by the refresh schedule.
- CKE must not fall during MRR, MRW, RD, or WR; it may fall after ACT/PRE/REF once their timing allows.
- Exit: CKE rises with CS_n high; first valid command after tXP = max(7.5 ns, 3 nCK). At least two clock transitions must occur during tXP.

8.11 Deep power-down

- Entry: CKE falls with the PRE code on the bus; all banks must be idle. Array contents are lost.
- The device must stay in DPD for at least tDPD = 500 us.
- Exit: CKE rises; the full power-up initialization sequence is required.

8.12 Clock stop and frequency change

Clock stop or frequency change is allowed only when CKE is low and no ACT/PRE/RD/WR/MRW/MRR are in flight, or when CKE is high with CS_n high and all related timings (tRCD, tRP, tWR, tMRW, etc.) are satisfied. After a CKE-high change or stop, normal operation resumes only after the clock is stable for at least 2 x tCK + tXP.

8.13 NOP / Deselect

Holds the bus idle and never aborts an in-progress operation. It is the only command legal during most power-state transitions and during tRFC/tMRW/tMRR windows.

Source: JESD209-3C sections 4.1-4.6, 4.7-4.8, 4.9, 4.10-4.11, 4.12, 4.13-4.16

9 Walkthrough: AXI Read, Row Already Open

Configuration used in all Chapter 4 traces (the drill model): BL = 8, RL = 6, WL = 3, flat 8-bank device. Banks B0-B7, rows R0-R7, columns C0-C7.

9.1 Scenario

An AXI4 master issues a read that maps to bank B3, row R5, column C2. Row R5 is already open in B3, so the access is a row hit.

9.2 Command trace

```
RD  B3 C2          ; row already open - issue the burst immediately
NOP                ; tCCD = 4 clocks minimum between column commands
NOP                ; read data returns after RL*tCK + tDQSCK
NOP                ; DQS may arrive over a wide window because there is
no DLL
```

9.3 Why each command

- `RD B3 C2` - the row is open, so no ACT is needed. CA2r=H selects a read; C2 is the burst start column; AP=0 (CA0f) so the row stays open.
- LPDDR3 uses a fixed burst length of 8. There is no burst-chop option; every RD delivers eight contiguous words from the array.
- The first DQS edge appears RL clocks after the command, plus the tDQSCK window. Because LPDDR3 has no DLL, tDQSCK is a loose min-to-max range that can span more than one clock. The controller must sample data relative to the strobe, not to a fixed CK edge.
- No PRE: leaving the row open is the open-page policy; a later access to R5 in B3 is again a hit.

9.4 AXI view

- AR channel: one burst request. With a 32-bit DQ bus, BL8 covers 32 bytes in one DRAM burst.
- A wider AXI transaction would be split into multiple RD commands spaced tCCD apart, e.g. `RD B3 C2`, then `RD B3 C4` four clocks later.

9.5 What to notice

- A row hit costs exactly one CA cycle on the command bus. Everything else (tCCD, turnaround) is about scheduling the next access around it.
- The controller tracks one open row per bank; that bookkeeping determines whether an AXI request becomes a hit or a miss.

Source: JESD209-3C sections 3, 4.2, 4.4, 4.17.1 (derived walkthrough)

10 Walkthrough: AXI Read, Wrong Row Open

10.1 Scenario

An AXI4 read request maps to bank B3, row R6, column C0 - but row R5 is currently open in B3. This is a row conflict (miss): the wrong row must be closed before the requested row can be opened.

10.2 Command trace

```
PREpb B3          ; close the wrong row (tRAS from its ACT already met)
--- (tRPpb) ---   ; single-bank precharge time: bank unavailable
ACT B3 R6         ; open the requested row
--- (tRCD) ---    ; row-to-column delay
RD  B3 C0         ; burst read, AP=0, row stays open
NOP
NOP
NOP
```

10.3 Why each command

- `PREpb B3` - a bank holds exactly one open row. AB=0, so only B3 is precharged; every other bank is untouched. This is the per-bank precharge path, which uses the shorter tRPpb recovery.
- `--- (tRPpb) ---` - single-bank precharge time. A PREpb to one bank uses tRPpb; a PREab would cost tRPab and close banks that might still be doing useful work.
- `ACT B3 R6` - opens the requested row. The ACT-to-ACT spacing to B3 (prior ACT -> PREpb -> this ACT) satisfies tRC = tRAS + tRPpb by construction if tRAS was met before the PREpb.
- `--- (tRCD) ---` - row-to-column delay; the column circuitry cannot be driven until the row's data is in the sense amps.
- `RD B3 C0` - the read itself, identical to the hit case once the row is open.

10.4 AXI view

Same single AR request as the hit walkthrough - but the latency the master sees is now tRPpb + tRCD longer. Nothing on the AXI side changes; the penalty is entirely in DRAM scheduling.

10.5 What to notice

- Miss cost = tRPpb + tRCD before the first data can even start its RL journey.
- If the previous access to B3 had used AP=1, the bank would already be idle and only tRCD would be owed - that is the close-page trade in one line.
- PREpb may not be issued earlier than tRAS after the ACT that opened R5, nor earlier than tRTP after a read from R5. A scheduler that wants the earliest PREpb starts the tRAS/tRTP counters at the ACT/RD, not at the miss.

Source: JESD209-3C sections 4.1, 4.4, 4.7, 4.17.1 (derived walkthrough)

11 Walkthrough: AXI Write to an Idle Bank

11.1 Scenario

An AXI4 write request maps to bank B1, row R2, column C0. B1 is idle (no open row).

11.2 Command trace

```
ACT B1 R2          ; open the row (bank was idle - no PRE needed)
--- (tRCD) ---     ; row-to-column delay
WR  B1 C0          ; burst write, AP=0; write data starts WL*tCK + tDQSS
later
NOP                ; burst occupies the DQ bus; tWR recovery runs from
NOP                ; the last data, then the bank may be precharged
NOP
NOP
NOP
NOP
NOP
```

11.3 Why each command

- `ACT B1 R2` - an idle bank needs no precharge; ACT is legal as long as tRRD from the previous ACT (any bank) and the tFAW window (max 4 ACTs in a rolling tFAW on 8-bank parts) are met.
- `--- (tRCD) ---` - same row-to-column delay as a read.
- `WR B1 C0` - CA2r=L selects a write. The controller drives DQS with a tWPRE preamble and the first data WL clocks (plus tDQSS) after the command. Each beat must meet tDS/tDH around the DQS edge; DM pins mask any beats that should not be written.
- The trailing NOPs cover tWR (write recovery from the end of the burst data): the bank may not be precharged until tWR has elapsed, because the write is still being committed to the array.

11.4 AXI view

- AW and W channels carry the request; the DRAM does not see them separately. The controller issues WR only when it has (or is about to have) the write data, because LPDDR3 gives it no way to stall mid-burst.
- B (write response) can be returned once the data is accepted by the DRAM schedule; the tWR window is a DRAM-internal matter.

11.5 What to notice

- An idle-bank write miss costs tRCD, half the read-miss penalty - no tRP because nothing was open.
- Alternative: issue `WR B1 C0` with AP=1. The device then auto-precharges after the burst, and the bank returns to idle with no explicit PREpb - at the price of having programmed nWR = RU(tWR/tCK) correctly in MR1.
- tWR is the write-side analog of tRTP on the read side: both gate the PRE; both exist because the burst's last data and the array's precharge cannot overlap.

Source: JESD209-3C sections 4.1, 4.5, 4.7, 4.17.1 (derived walkthrough)

12 Walkthrough: Mixed Reads and Writes

12.1 Scenario

Three AXI requests arrive close together:

1. Read: bank B3, row R5 (open), column C2.
2. Write: bank B1, row R2 (open), column C0.
3. Read: bank B1, row R2, column C4.

Both rows are already open, so this trace is pure bus scheduling: the DQ bus is shared, and read/write direction changes cost bubbles.

12.2 Command trace

```
RD  B3 C2         ; read burst 1: data out after RL + tDQSCK
--- (tRTW) ---    ; read-to-write turnaround, see below
WR  B1 C0         ; write burst: DQ direction flips
--- (tWTR) ---    ; write-to-read recovery before any new RD
RD  B1 C4         ; read burst 2, same open row as the write - a hit
PREpb B1          ; done with B1 (tRTP after the RD met); close it
```

12.3 Why each gap

- Read-to-write turnaround (book symbol tRTW): the write command may be issued no earlier than RL + RU(tDQSCK(MAX)/tCK) + BL/2 + 1 - WL clocks after the read command. With RL=6, WL=3, BL=8 and tDQSCK(MAX)=2 clocks, that is 6 + 2 + 4 + 1 - 3 = 10 clocks. The bubble exists because read data is still leaving the DRAM and the DQ bus must be idle before the controller starts driving write data. LPDDR3 has no DLL, so tDQSCK is a wide window rather than a locked edge; the controller must budget the worst-case tDQSCK(MAX).
- Write-to-read: the next read command waits WL + 1 + BL/2 + RU(tWTR/tCK) clocks after the write command (3 + 1 + 4 + 4 = 12 clocks here). tWTR lets the write data finish propagating into the array before sense amps are asked to read.
- `RD B1 C4` - same bank, same row as the write: a row hit even though a write happened in between. Open-row state is per bank, not per access type.
- `PREpb B1` - legal once tRAS (from the ACT that opened R2) and tRTP (from RD B1 C4) are both met. A PREab would have cost tRPab and closed every bank.

12.4 AXI view

AXI is oblivious to direction changes; its read and write channels are independent. The DRAM bus is not: every RD->WR or WR->RD transition on the shared DQ pins costs turnaround clocks, so a controller that groups reads with reads and writes with writes (batching) measurably outruns one that serves requests strictly in arrival order.

12.5 What to notice

- Turnarounds, not tRCD, dominate lightly-loaded mixed traffic.
- tCCD still applies between column commands: even without the direction change, bursts on the shared bus must not collide. With BL=8, tCCD is 4 clocks.
- tFAW limits activate parallelism: no more than four ACT or REFpb commands may fall inside a rolling tFAW window on this 8-bank device.
- Refresh appears as REFab in most schedules, but REFpb exists as a per-bank alternative that refreshes one bank while others remain active.

Source: JESD209-3C sections 4.4, 4.5, 4.7, 4.8, 4.17.1 (derived walkthrough)

13 Data Path

13.1 Prefetch and burst

LPDDR3 uses an 8n prefetch: each column access moves eight bits per DQ between the array and the IO, serialized as four UI on each of two clock cycles. The fixed burst length is BL8. There is no burst chop and no interleaved burst type; every column transaction delivers eight consecutive column locations starting from the commanded column address.

13.2 Read and write strobes

A single bidirectional differential pair (DQS_t/DQS_c) serves each byte lane.

Direction	Driver	Alignment	Preamble	Postamble
Read	DRAM	Edge-aligned to DQ	tRPRE >= 0.9 tCK	tRPST >= 0.3 tCK
Write	Controller	Centered in DQ eye	tWPRE >= 0.8 tCK	tWPST >= 0.4 tCK

There is no DLL. The read strobe is timed with a wide access window rather than a locked phase: tDQSCK ranges from 2.5 ns (min) to 5.5 ns (max). The controller must train its capture logic to absorb this spread, using the MR32/MR40 DQ calibration patterns and the normal read eye.

13.3 Read and write latencies

RL and WL are programmed through MR2. Two WL sets (A and B) trade extra write latency for easier system timing at the higher speeds.

MR2 OP<3:0>	RL	WL Set A	WL Set B	Max CK freq
0001B	3	1	1	<= 166 MHz (optional)
0100B	6	3	3	<= 400 MHz
0110B	8	4	4	<= 533 MHz
0111B	9	5	5	<= 600 MHz
1000B	10	6	8	<= 667 MHz (default)
1001B	11	6	9	<= 733 MHz
1010B	12	6	9	<= 800 MHz
1100B	14	8	11	<= 933 MHz
1110B	16	8	13	<= 1066 MHz

Symbol	Timing relationship
tDQSCK	CK edge to first valid DQS edge: 2.5 ns min, 5.5 ns max
tDQSQ	DQS-to-DQ skew: max 165/135/115/100 ps at 1333/1600/1866/2133 MT/s
RD cmd -> first DQ	RL x tCK + tDQSCK + tDQSQ
WR cmd -> first DQS	WL x tCK + tDQSS
tDQSS	WR cmd CK edge to first DQS latching edge: 0.75-1.25 tCK
tDQSH / tDQSL	DQS high/low pulse width during writes: >= 0.4 tCK
tDSS / tDSH	DQS falling edge setup/hold to CK: >= 0.2 tCK
tDS / tDH	DQ/DM setup/hold to DQS: 175/150/130/115 ps at

Symbol	Timing relationship
	1333/1600/1866/2133

Symbol	Turnaround relationship
tCCD	CAS-to-CAS delay: 4 tCK
tWTR	WR -> RD: max(7.5 ns, 4 tCK), measured from the CK edge after the last write datum
tRTW (book symbol)	RD -> WR: RL + RU(tDQSCK(MAX)/tCK) + BL/2 + 1 - WL clock cycles. JESD209-3C does not name this symbol; it is the cross-book convention.

13.4 DM - write data mask

One DM pin per byte lane is sampled on both DQS edges with the write data. When DM is high for a beat, that byte is masked and not written. DM loading matches DQ/DQS loading so timing tracks the data. Partial writes (e.g. AXI W with sparse WSTRB) can use DM instead of a read-modify-write.

13.5 What LPDDR3 does not have

- No data bus inversion (DBI) on reads or writes.
- No parity on the CA bus and no CRC on the data bus.
- No on-die ECC.
- No burst chop, no interleaved burst order, and no additive latency.

Every DQ cycle carries payload; the controller must provide any error protection.

13.6 Signaling

The DQ/DQS/DM interface is HSUL_12: 1.2 V VDDQ (nominal 1.20 V, min 1.14 V, max 1.30 V), with VREFDQ near VDDQ/2 at the receiver. Unlike LPDDR2, LPDDR3 adds a dedicated ODT pin and programmable on-die termination for the data bus; termination is asynchronous and is covered in the next file.

Source: JESD209-3C sections 3.4.1 (MR1, MR2), 4.3, 4.4, 4.5, 4.6, 4.10.2, 4.11.2, 8.6.1, Table 64

14 On-Die Termination

LPDDR3 exposes a dedicated ODT pin that turns termination on or off for each DQ, DQS_t, DQS_c, and DM line. The pin is not sampled by the clock; the decision is asynchronous. Termination is intended to improve channel signal integrity and is controlled independently per device by the memory controller.

14.1 Enabling and selecting the termination value

ODT is configured through MR11.

MR11 OP<1:0>	RTT value	Notes
00B	Disabled	Default
01B	RZQ/4 = 60 ohm	Required for 1866/2133; optional for 1333/1600
10B	RZQ/2 = 120 ohm	
11B	RZQ/1 = 240 ohm	

MR11 OP<2> selects whether ODT remains functional during CKE power-down: 0B disables it in power-down (default); 1B leaves it enabled, in which case VDDQ must stay on during power-down.

The absolute accuracy of the on-die termination is maintained by ZQ calibration (MRW ZQ init, long, or short) against the external 240 ohm RZQ resistor.

14.2 Asynchronous ODT timing

Because the ODT pin is asynchronous, the only pin-controlled delays are the pure analog turn-on and turn-off windows.

Symbol	Definition	Window
tODTon	ODT pin high to RTT beginning to turn on / fully on	1.75 ns min to 3.5 ns max
tODToff	ODT pin low to RTT beginning to turn off / at high-Z	1.75 ns min to 3.5 ns max

14.3 ODT during reads

The DRAM automatically disables termination during any read (RD or MRR) and re-enables it after the read data completes, even if the ODT pin stays asserted. The automatic timing is referenced from the read-data clocks:

Symbol	Definition
tAODToff	Automatic RTT turn-off after READ data: min = tDQSCK,min - 300 ps, referenced from RL - 2 CK
tAODTon	Automatic RTT turn-on after READ data: max = tDQSCK + 1.4 x tDQSQ,max + tCK(avg,min), referenced from RL + BL/2 CK

14.4 ODT during low-power states and training

State	ODT behavior
CKE power-down, OP<2>=0	Disabled within tODTd (max 12 ns); re-enabled within tODTe (max 12 ns)
Self-refresh entry	Disabled within tODTd (max 12 ns + 0.5 tCK)
Self-refresh exit	Re-enabled within tODTe (max 12 ns)
Deep power-down	Disabled within tODTd; full re-init required on exit
CA training	Disabled; ODT pin ignored
Write leveling	ODT pin high -> DQS_t/DQS_c termination ON, DQ termination OFF; ODT must be high if enabled

14.5 ODT states truth table

Mode	DQ termination	DQS termination
Write	Enabled (if MR11 enabled and ODT HIGH)	Enabled
Read / DQ calibration	Disabled	Disabled
ZQ calibration	Disabled	Disabled
CA training	Disabled	Disabled
Write leveling, ODT asserted	Disabled	Enabled

14.6 What LPDDR3 does not have

- No synchronous ODT latency (no ODTL-on/off as in DDR3/DDR4).
- No VTT termination rail; termination is purely on-die to VDDQ.
- No DBI, no parity, no CRC, and no on-die ECC.

Source: JESD209-3C sections 4.12, 4.12.1-4.12.7, Table 22, Table 23, Table 54, Table 64

15 Core Timing Parameters

The bank-state timings: what each parameter constrains, and the command transition it applies to. Values are from the AC timing table (Table 64); clock conversion is divide by tCK(avg) and round up (RU).

15.1 Row-cycle parameters

LPDDR3 carries TWO precharge times - one per-bank, one all-bank - and the choice follows the command, not the register:

Symbol	Definition	Applies between	Min
tRCD	Row-to-column delay; three speed grades exist (fast/typ/slow)	ACT(bank x) -> RD/WR(bank x)	max(15/18/24 ns, 3 nCK)
tRPpb	Precharge recovery, single bank	PRE(bank x) -> ACT(bank x)	max(15/18/24 ns, 3 nCK)
tRPpab	Precharge recovery, all banks	PREA -> ACT(any)	max(18/21/27 ns, 3 nCK)
tRAS	Minimum (and maximum) row-open time	ACT(bank x) -> PRE(bank x)	min max(42 ns, 3 nCK); max min(70.2 us, 9 x RM x tREFI)
tRC	Full row cycle; built from whichever precharge applies	ACT(bank x) -> ACT(bank x)	tRAS + tRPpab, or tRAS + tRPpb

The fast/typ/slow tRCD and tRP grades are a low-power knob: a slower setting eases the analog timing budget (and saves power) at the cost of latency. The tRAS maximum is where the refresh multiplier RM (MR4) shows up in scheduling math.

15.2 Activate-rate parameters (cross-bank)

Symbol	Definition	Applies between	Min
tRRD	Activate-to-activate spacing, any two banks (limits peak array current)	ACT(bank x) -> ACT(bank y)	max(10 ns, 2 nCK)
tFAW	Four-activate rolling window: at most 4 ACTs (or REFpb) inside the window	any 4 ACTs within a sliding window	max(50 ns, 8 nCK)

The spec's own window example: if RU(tFAW/tCK) = 10 and an ACT lands in clock n, at most three more ACTs may land in clocks n+1 .. n+9. Note tFAW counts REFpb as well as ACT - per-bank refresh consumes activate budget.

15.3 Data-to-precharge parameters

Symbol	Definition	Applies between	Min
tRTP	Internal analog read must finish before the row closes	last read data(bank x) -> PRE(bank x)	max(7.5 ns, 4 nCK)
tWR	Write recovery: last write data committed to the cells before precharge	last write data(bank x) -> PRE(bank x)	max(15 ns, 4 nCK)

Command-spacing formulas that fall out of these (same bank):

Transition	Earliest spacing (clocks)
RD -> PRE	BL/2 + max(4, RU(tRTP/tCK)) - 4
WR -> PRE	WL + BL/2 + RU(tWR/tCK) + 1
RDA/WRA -> ACT	precharge fires automatically after the recovery window; next ACT >= tRP after it actually starts, and >= tRC from the old ACT

15.4 Why these numbers dominate

tRCD and tRP are the row-miss penalty - and LPDDR3 makes the controller choose per precharge whether it pays tRPpb or the larger tRPpab. tRAS + tRP = tRC bounds per-bank row cycling; tRRD and tFAW bound device- wide activate rate, with tFAW also metering per-bank refreshes. The scheduler hides tRCD/tRPpb behind other banks' data and paces ACTs (and REFpBs) against tRRD/tFAW.

Source: JESD209-3C sections 4.1, 4.7, 11.4 (Table 64)

16 Turnaround and Burst Timing

Column-command spacing and bus-direction rules. These decide how full the DQ bus can be kept.

16.1 Latencies and CAS-to-CAS

LPDDR3 has no DLL and no additive latency: read and write latency are programmed directly (MR0 OP6/OP7 select the WL set and the optional RL=3/WL=1 low-latency pair) and the latencies scale with clock speed.

Symbol	Definition	Applies between	Value
tCCD	Column command to column command, same or different banks	RD/WR -> RD/WR (same direction)	4 clocks
RL	Read latency: command to first data	RD -> first DQS/DQ out	programmed: 3-16 clocks by rate (6/8/9/10/11/12/14/16 at 800-2133 MT/s)
WL	Write latency: command to first strobe	WR -> first DQS in	programmed set A: 1/3/4/5/6/8; set B (optional): up to 13

BL8 occupies 4 clocks of DQ. Seamless same-direction traffic issues a column command every BL/2 = 4 clocks, which is exactly tCCD - back-to-back BL8 bursts just saturate the bus. There is no burst chop: BL8 is the only burst length, and no interleaved burst type exists.

16.2 Direction changes

Note on tRTW: JESD209-3C never names the read-to-write parameter (the symbol does not appear in the spec). It does specify the spacing, and the spacing depends on tDQSCK because without a DLL the read strobe can lag the clock by more than a clock. This book calls the parameter tRTW, matching the other books in this series.

Symbol	Definition	Applies between	Value
tRTW	Read-to-write turnaround: the loose read strobe (tDQSCK) plus burst must clear before the write preamble (book symbol; the spec gives the relation unnamed)	RD(any bank) -> WR(any bank)	RL + RU(tDQSCK(MAX)/tCK) + BL/2 + 1 - WL
tWTR	Write-to-read: last write data must clear the input path before a read reuses the array	last write data(any) -> RD(any)	max(7.5 ns, 4 nCK)

Command-spacing forms:

Transition	Spacing from first command (clocks)
RD -> WR	RL + RU(tDQSCK(MAX)/tCK) + BL/2 + 1 - WL
WR -> RD	WL + 1 + BL/2 + RU(tWTR/tCK)
RD -> RD	max(tCCD, BL/2)
WR -> WR	max(tCCD, BL/2)

Worked example (the Chapter 4 configuration: RL = 6, WL = 3, BL8, tWTR = 4 clk, tDQSCK = 2 clk): RD->WR = 6 + 2 + 4 + 1 - 3 = 10 clocks, WR->RD = 3 + 1 + 4 + 4 = 12 clocks. The asymmetry is the write path's long internal tail; reads turn into writes cheaper than writes turn into reads. The tDQSCK term is the LPDDR3 signature: a DLL-based DRAM replaces it with a locked value, an LPDDR3 controller must budget the whole min/max window.

16.3 Strobe-level timing (for completeness)

These bound the DQS/DQ relationship; the PHY must meet them. With no DLL, several are wide windows rather than tight skews:

Symbol	Definition	Value
tDQSCK	Read DQS access from CK_t/CK_c: a 2.5-5.5 ns window (not a locked skew)	min 2500 ps, max 5500 ps; 5620 ps derated
tDQSCKDS/DM/DL	DQSCK drift across 160 ns / 1.6 us / 32 ms rolling windows (guaranteed by design, untested)	265/593/733 ps max at 1333, shrinking with rate
tDQSQ	DQ skew against DQS, per group	165 ps max at 1333, down to 100 ps at 2133
tQH	DQ/DQS output hold	min(tQSH, tQSL)
tDQSS	Write DQS first latching edge to CK	0.75 to 1.25 tCK (wide - no DLL)
tWPRE / tWPST	Write DQS preamble / postamble	0.8 / 0.4 tCK
tRPRE / tRPST	Read DQS preamble / postamble	0.9 / 0.3 tCK
tDS / tDH	Write DQ setup / hold against DQS	175 ps min at 1333, down to 115 ps at 2133
tDQSH / tDQSL	Write DQS input high/low pulse width	0.4 tCK

16.4 Temperature derating

Above the 85 C threshold the core analog timings stretch: tRCD, tRC, tRAS, tRP and tRRD each carry a +1.875 ns derated minimum, and tDQSCK's max becomes 5620 ps. A controller that runs hot must add the derating on top of the programmed values rather than discovering it on the bus.

Source: JESD209-3C sections 4.4, 4.5, 11.4 (Table 64)

17 Refresh and Self-Refresh Timing

17.1 Refresh parameters

LPDDR3 counts refresh the LPDDR way: a fixed number of REFRESH commands per refresh window, at either all-bank or per-bank granularity.

Symbol	Definition	Value
tREFW	Refresh window: every row refreshed inside it	32 ms (16 ms at 1/2-rate, 8 ms at 1/4-rate PASR settings)
R	Required REFab commands per window	4096 (8192 at 8 Gb and above)
tREFI	Average interval, REFab	7.8 us
tREFIpb	Average interval, REFpb (per bank)	0.975 us at 1 Gb, 0.4875 us above (8 x tREFIpb covers all banks)
tRFCab	All-bank refresh cycle time	130 ns (1-4 Gb), 210 ns (6-8 Gb)
tRFCpb	Per-bank refresh cycle time	60 ns (1-4 Gb), 90 ns (6-8 Gb)

Rules:

- All banks must be idle (and tRP met) before REFab; REFpb only needs its target bank idle, and the target walks banks round-robin via an internal counter.
- REFpb buys scheduling freedom: one bank restores while the others keep serving. Eight REFpb (one per bank) equals one REFab of coverage.
- tFAW counts REFpb as an activate-class command - a cluster of per-bank refreshes eats activate budget.
- The refresh rate is set through MR4 (full, 1/2, 1/4 rate; temperature derating flag), and the RM multiplier in the tRAS maximum is the same knob seen from the row-timing side.
- Violating the refresh window silently corrupts data.

At 8 Gb, REFab at 210 ns per 7.8 us is about 2.7% of bus time; per-bank refresh cuts the all-bank stall to 60-90 ns per bank.

17.2 Self-refresh

The DRAM refreshes itself with the external clock stopped: the mobile sleep state. PASR (MR16 bank mask, MR17 segment mask) restricts refresh to a fraction of the array; anything outside is lost on entry.

Symbol	Definition	Value
tCKE	Minimum CKE pulse width, high or low	max(7.5 ns, 3 nCK)
tCKESR	Minimum CKE low width for self-refresh entry to exit	max(15 ns, 3 nCK)
tXSR	Exit self-refresh to next valid command	max(tRFCab + 10 ns, 2 nCK)
tCKSRE/tCKSRX	Valid clock required after entry / before exit	max(5 nCK, 10 ns) class (clock-stop rules)

There is no separate DLL-relock wait as in DDR3: LPDDR3 has no DLL. Exit needs a stable clock, CKE high, then tXSR. The temperature sensor (MR4) doubles the internal refresh rate when the die runs hot.

17.3 Power-down and deep power-down

Symbol	Definition	Value
tXP	Exit power-down to any valid command	max(7.5 ns, 3 nCK)
tMRRI	After tXP, extra wait before an MRR	tRCD(min)
tCPDED	Command pass disable delay after PD entry	2 tCK
tDPD	Minimum deep power-down time	500 us

Power-down keeps the array alive (refresh obligations continue: 9 x tREFI of slack at best); deep power-down does not - DPD drops the array contents and every ODT/input for minimum power, and exit is a complete re-initialization. ODT timing around the power states: disabled within 12 ns of PD entry, 12 + 0.5 tCK of SR/DPD entry, and re-enabled within 12 ns of exit. Clock stop is legal in idle and power-down, which is the other big LPDDR3 power knob alongside DPD.

Source: JESD209-3C sections 4.8, 4.9, 4.13, 4.14, 11.3 (Table 62), 11.4 (Table 64)

18 Package and Electrical Notes

A light pass: what matters when connecting an LPDDR3 device, deliberately skipping the full ballout grids (consult the spec figures for those).

18.1 Packages

- Two delivery forms: package-on-package (PoP, stacked above the application processor) and discrete FBGA; both use the same signal definitions with per-form ballouts in sec 2.1 and 2.2.
- Organizations are x16 and x32 on a single channel; 8 banks flat, no bank groups. Densities run 1 Gb to 32 Gb (the addressing table carries 6 Gb and 12 Gb steps alongside the powers of two).
- No RESET# pin: reset is a command (MRW to the reset register). There IS a dedicated ZQ pin bonding to the external 240 ohm reference resistor.

18.2 Supplies and signaling

Rail	Nominal	Feeds
VDD1	1.2 V class	core/array
VDD2	1.14-1.30 V	core power
VDDQ	1.14-1.30 V	I/O buffers
VREFCA / VREFDQ	0.5 x VDDQ	input references

- HSUL_12 (High-Speed Unterminated Logic, 1.2 V): the bus is unterminated by design - no VTT rail, no board termination. Signal integrity comes from controlled drive strength (MR-programmed, ZQ- calibrated) plus the on-die termination for DQ, not from a terminated channel.
- CA bus: single-clock-rate command/address on a narrow, shared CA pin group (commands span two cycles); CK_t/CK_c differential; DQS_t/DQS_c differential per byte group, driven by the DRAM on reads and by the controller on writes.

18.3 Speed grades

Per the latency table (Table 63): data rates 333 through 2133 MT/s (tCK 6 ns down to 0.938 ns), with the programmed RL/WL pair tracking the rate - RL 6 to 16, WL 3 to 8 (set A) across the mainstream range, plus the optional RL=3/WL=1 low-latency pair and WL set B. Boot operation runs from a 10-55 MHz clock (tCKb 18-100 ns) during initialization.

18.4 Derating and characterization

Above 85 C the core timings take +1.875 ns derating and tDQSCK stretches to 5620 ps max (see the turnaround chapter). Section 10 carries the IDD measurement conditions (per-state current tables, the mobile power budget); section 9 the pin capacitance budgets. Neither changes scheduling; consult them when sizing the power delivery.

Source: JESD209-3C sections 2, 6.1, 11.3 (Table 63)

19 Test Features

LPDDR3 has no JTAG boundary scan, no DDR3-style MPR, no OCD calibration engine, and no standard write-CRC or gear-down mode. Instead it exposes a small set of calibration and observation mechanisms, all reached through MRW or MRR.

19.1 MRR as the observation port

A Mode Register Read returns the contents of any mode register. The command is recognized when CS_n is low with CA0-CA2 low and CA3 high; the register address is built from the remaining CA fields. Valid data appears on DQ[7:0] on the first beat after the programmed read latency, and the burst is always eight clocks long. Only NOP is legal during the tMRR interval. MRR is the normal way to read MR0 at boot to confirm device-auto-initialization status, ZQ pin self-test results, and supported write-latency/read-latency options.

19.2 DQ calibration (MR32 / MR40)

Because LPDDR3 has no DLL, the controller must train its read capture against a known pattern. MRR to MR32 returns Pattern A and MRR to MR40 returns Pattern B. For x16 parts the expected bits are driven on DQ0 and DQ8; for x32 parts they are driven on DQ0, DQ8, DQ16, and DQ24. The remaining data pins may either repeat DQ0 or drive zero. Pattern A toggles every bit time (1-0-1-0…), while Pattern B produces pairs of identical values (0-0-1-1-0-0-1-1). Unlike ordinary MRR bursts, the DQ calibration bursts define valid content on every beat so the host can tune capture timing on multiple edges.

19.3 ZQ calibration (MR10)

Output-driver and ODT calibration is initiated by MRW to MR10 with one of four opcodes:

Opcode	Command	Typical use
0xFF	ZQINIT	Required once after reset; ~1 us
0xAB	ZQCL	Long maintenance calibration; ~360 ns
0x56	ZQCS	Short maintenance calibration; ~90 ns
0xC3	ZQRESET	Return to default calibration; max(50 ns, 3 tCK)

A 240 ohm, +/-1% resistor must be connected between the ZQ ball and ground. ZQ commands are only legal when all banks are precharged, ODT is disabled, and the DQ bus is quiet; only NOP may be issued during the calibration window. If several devices share one ZQ resistor, their initialization, long, and short calibrations must not overlap, although ZQRESET overlap is allowed.

19.4 CA training (MR41 / MR48 / MR42)

Commands travel across the narrow CA bus over two cycles, so the controller must align each CA bit to the clock before normal operation. Training is entered by MRW to MR41, performed in two passes, and exited by MRW to MR42.

- First pass (MR41): calibrate CA0, CA1, CA2, CA3, CA5, CA6, CA7, and CA8. The DRAM echoes each CA value on specific DQ pins according to the MR41 CA-to-DQ map.
- Mapping change (MR48): switch the mapping so the remaining pins can be observed.
- Second pass (MR48): calibrate CA4 and CA9.
- Exit (MR42): return to normal operation.

MR41, MR48, and MR42 use special encodings that are intentionally identical on the rising and falling clock edges so the DRAM can recognize them even before CA timing is adjusted. During training the clock phase may be intentionally varied while CS_n is high and CKE is low.

Key CA-training timings:

Symbol	Definition	Value
tCAMRD	First CA calibration command after mode entry	20 tCK
tCAENT	First CA calibration command after CKE goes LOW	10 tCK
tCACKEL	CKE LOW after CA-training mode is programmed	10 tCK
tCACKEH	CKE HIGH after the last CA-training result is driven	10 tCK
tCAEXT	CA-training exit command after CKE is HIGH	10 tCK
tCACD	Delay from one CA calibration command to the next	RU(tADR + 2 x tCK)

Symbol	Definition	Value
tADR	Data-out delay after a calibration command	20 ns max
tMRZ	MRW CA-training exit command to DQ tri-state	3 ns min

19.5 Write leveling (MR2[7])

Write leveling lets the controller align each DQS rising edge to the clock at the DRAM pin. The device enters write-leveling mode when MR2[7] is set HIGH and leaves when MR2[7] is cleared. Only NOP or the exit MRW is legal while in this mode.

The controller first holds DQS_t LOW and DQS_c HIGH for at least tWLDQSEN, then issues the first DQS edge no sooner than tWLMRD. The DRAM samples CK with that DQS rising edge and asynchronously drives the sampled value onto every DQ bit in the byte after tWLO. The controller iterates its DQS delay until it sees a 0-to-1 transition, which locks the DQS-to-CK relationship needed for tDQSS.

Symbol	Definition	Value
tWLDQSEN	DQS_t/DQS_c delay after entering write-leveling mode	25 ns min
tWLMRD	First DQS edge after entering write-leveling mode	40 ns min
tWLO	Write-leveling output delay	0 to 20 ns
tWLS	DQS rising-edge setup to CK	135-205 ps
tWLH	DQS rising-edge hold from CK	135-205 ps

19.6 Temperature sensor (MR4)

The on-die temperature sensor is read through MR4. The lower three bits report a refresh-rate multiplier that ranges from 4x down to 0.25x and also indicate whether the die is below the low-temperature limit or above the high-temperature limit. Bit 7 is the Temperature Update Flag (TUF), which is set whenever the 3-bit value has changed since the last MR4 read; reading MR4 clears TUF. The device updates the sensor at least every tTSI = 32 ms, and the value is never older than tTSI after exiting self-refresh or power-down.

19.7 What is not here

- No DDR3 MPR; LPDDR3 uses MRR and the DQ calibration registers instead.
- No standard boundary scan on the device.
- No OCD engine; ZQ calibration replaces it.
- No write CRC, no gear-down mode, and no per-pin loopback.
- Vendor test modes exist only as a reserved write-only register (MR9) with vendor-defined behavior.

Source: JESD209-3C sections 4.10, 4.10.1, 4.10.2, 4.11.2, 4.11.3, 4.11.4, 3.4.1